

GrandStay Hotels & Resorts

Database Design, Implementation and Population

Document Type	SQL Database Design & Implementation Assignment
Reference	DWD-SQL-2026-001
Program	Data with Danny — Cohort 8
Tool	SQL Server (SSMS)
Deadline	Wednesday — 2 weeks from issue date
Submission	datawithdany@gmail.com

01 BACKGROUND

You have been brought in as a database consultant for GrandStay Hotels & Resorts, a growing hospitality group operating properties across six international cities. The business has been running its full operations on a single shared Excel workbook for the past four years. That workbook now has over 50 columns, duplicate guest records, broken formulas, comma-separated service fields, and no reliable way to answer basic business questions. Your job is to fix that — permanently.

GrandStay operates properties in Lagos, Abuja, London, Dubai, Accra, and Nairobi. Each property handles its own bookings, room assignments, guest check-ins, ancillary service orders, and payment collections. All of this flows into one Excel file maintained by the operations team. The CEO has mandated a move to a proper relational database before the next financial year.

02 THE PROBLEM

The supply chain analytics team needs a model that can answer the following six business questions. Your data model must be capable of answering all six. If it cannot, it is incomplete.

QUESTIONS YOUR DATABASE MUST BE ABLE TO ANSWER

- Which hotels generate the most revenue and what is the average revenue per booking?
- Which room types have the highest occupancy rates across all properties?
- Which guest loyalty tier spends the most on average per stay?
- What is the most popular ancillary service category by total revenue?
- Which bookings currently have an outstanding balance due?
- How does booking volume and revenue trend month by month?

03 OBJECTIVE

Using SQL Server and SSMS, design and implement a fully normalised relational database that replaces the Excel workbook above. You must make every structural decision yourself — there is no template to follow. Your database must be capable of answering all six business questions above using SQL queries that JOIN across multiple tables.

04 THE EXCEL FILE — COLUMN DICTIONARY

Below is the full column dictionary of the current Excel workbook — every column the operations team currently uses. Study it carefully before designing anything. Your job is to take this flat, messy structure and transform it into a properly normalised relational database.

THE DATA HAS ISSUES

Mixed date formats. Duplicate records. Comma-separated multi-value fields. Redundant calculated columns. Free-text fields with no standards. That is intentional — real data is never clean.

Column Name	Example Value	Notes / Issues
booking_ref	BK-2024-00451	Free-text booking reference. Not always unique — duplicates exist.
booking_date	14/03/2024	Date booking was made. Mixed formats (DD/MM/YYYY and MM-DD-YYYY).
check_in_date	20/03/2024	Scheduled check-in date.
check_out_date	24/03/2024	Scheduled check-out date.
actual_check_in	20/03/2024 14:35	Actual arrival datetime. Blank if guest did not show.
actual_check_out	24/03/2024 11:10	Actual departure datetime. Blank if still checked in.
nights_stayed	4	Calculated field (=checkout - checkin). Unreliable when dates conflict.
booking_status	Confirmed	Values: Confirmed, Checked In, Checked Out, Cancelled, No Show.
booking_channel	Direct Website	Direct Website, OTA, Walk-In, Corporate, Travel Agent.
hotel_name	GrandStay Lagos Victoria	Full hotel name. Inconsistent spelling across rows.
hotel_city	Lagos	City where hotel is located.
hotel_country	Nigeria	Country of the hotel.
hotel_star_rating	5	Star rating of the property (3, 4, or 5).
room_number	204	Physical room number. Must be combined with hotel to be unique.
room_type	Deluxe King	Standard, Superior, Deluxe, Suite, Presidential.
floor	2	Floor number of the room.
max_occupancy	2	Maximum number of guests the room holds.
bed_type	King	King, Queen, Twin, Single.
room_rate_per_night	45000	Nightly rate. Currency not stored — causes reporting problems.
total_room_charge	180000	room_rate x nights. Recalculated column — errors present.
guest_first_name	Adaeze	Guest first name.
guest_last_name	Okonkwo	Guest last name.
guest_email	adaeze.o@gmail.com	Primary contact. Duplicates exist for repeat guests.
guest_phone	+234 803 456 7890	No standard format enforced.
guest_nationality	Nigerian	Free-text. Not standardised — same country spelled multiple ways.
guest_dob	12/07/1988	Date of birth. Used for promotions. Often blank.

guest_id_type	Passport	Passport, National ID, Drivers Licence.
guest_id_number	A12345678	ID document number stored in plain text.
loyalty_tier	Gold	None, Silver, Gold, Platinum. Assigned manually — inconsistent.
loyalty_points_balance	4200	Current points. Not linked to transactions — manually updated.
number_of_guests	2	Number of guests on this booking.
total_room_revenue	180000	Duplicate of total_room_charge. Different column name, same data.
service_charges_total	35000	Sum of all services. Not itemised in this column.
discount_applied	10%	Stored as percentage text — cannot be used in calculations.
discount_amount	18000	Monetary discount. Sometimes conflicts with discount_applied.
tax_amount	19700	VAT or applicable tax on the booking.
grand_total	216700	Total due including services, discount, and tax.
amount_paid	150000	Total payments received. May be partial.
balance_due	66700	grand_total minus amount_paid. Recalculated — errors present.
payment_method	Card	Cash, Card, Bank Transfer, Mobile Money, Corporate Account.
payment_date	20/03/2024	Date of last payment only — does not support multiple payments.
payment_status	Partially Paid	Pending, Partially Paid, Paid, Refunded.
services_used	Spa, Airport Transfer, Minibar	Comma-separated list. Impossible to aggregate reliably.
service_dates	21/03, 20/03, 22/03	Corresponding dates per service. Comma-separated — fragile.
service_costs	15000, 12000, 8000	Cost per service. Comma-separated. No currency stored.
assigned_staff_name	Emeka Nwosu	Front desk staff who handled check-in. Free text.
special_requests	Late check-out, extra pillows	Free-text. Not standardised.
cancellation_reason	Change of travel plans	Free text. Only populated for cancelled bookings.
notes	VIP guest — handle with care	General free-text notes. No character limit enforced.

05 THE ASSIGNMENT

Step 1 — Analyse the Excel Dictionary

Read every column carefully before touching SQL. Identify which columns represent entities, which represent measurements, and which are derived or redundant. What does one row in this Excel file represent? That is your grain. Document your answer before designing anything — everything else flows from this.

Step 2 — Design the ER Diagram

Identify all the entities that exist in this business. For each entity, determine its attributes and its relationships to other entities. Draw a complete Entity Relationship Diagram showing all tables, primary keys, foreign keys, and cardinality. You may use any tool — Excalidraw, draw.io, Lucidchart, or hand-drawn and photographed. Your ERD must be included in your documentation.

Step 3 — Write the DDL Script

Implement your design in SQL Server. Your script must: create the database, create all tables in correct dependency order, define all primary keys using IDENTITY, name all foreign key constraints, enforce business rules with CHECK constraints, apply NOT NULL and DEFAULT values appropriately, include IsActive flags for soft delete, include CreatedAt and UpdatedAt audit timestamps, and create non-clustered indexes on foreign key and high-frequency filter columns.

Step 4 — Write the DML Script

Populate your database with realistic data. At minimum: 6 hotels, 5 room types, at least 30 rooms, at least 20 guests, at least 40 bookings with a mix of statuses, at least 15 service charge records, and invoices or payments with a mix of Paid, Partially Paid, and Overdue records. You must also include one example each of UPDATE (with a business justification), soft DELETE (IsActive = 0), hard DELETE inside a transaction, and MERGE (syncing from a staging table).

Step 5 — Write the Validation Queries

Write one SQL query for each of the six business questions in Section 02. Every query must JOIN across multiple tables. A query that returns results from a single table will not be accepted. Run each query, screenshot the result, and embed it in your documentation.

06 VALIDATION TESTS

You must run all six tests and include a screenshot of each result in your documentation. All six must pass. A model that cannot pass all six tests is not complete.

TEST	NAME	WHAT IT VALIDATES
01	Hotel Revenue Summary	Total revenue and average revenue per booking grouped by hotel. Validates Hotel and Booking tables are correctly joined.
02	Room Type Occupancy	Count of completed bookings by room type. Validates your Room and RoomType table connections.
03	Guest Loyalty Spend	Average total spend per booking grouped by loyalty tier. Validates your Guest table connection.
04	Service Category Revenue	Total revenue grouped by service category. Validates your Service and ServiceCharge tables.

05	Outstanding Balances	All bookings with a balance due greater than zero, ordered by balance descending. Validates your Payment table.
06	Monthly Booking Trend	Total bookings and total revenue by month and year, ordered chronologically. Validates date handling across the full model.

07 DOCUMENTATION REQUIREMENTS

Every student must submit a written document alongside their SQL files. Your documentation is not optional. It is half the assignment. Marks will be deducted for incomplete or vague documentation.

- How you interpreted the grain of the data from the Excel column dictionary
- Which Excel columns you kept, transformed, and discarded — and why for each decision
- How you identified each entity and what made it a separate table
- How you handled the data quality issues present in the Excel file
- What primary key strategy you used and why
- How you decided which columns belong in which table
- Screenshots of all six validation queries returning actual results
- One thing you would do differently if you had more time

08 GRADING CRITERIA

Your submission will be assessed on four criteria:

CRITERIA	DESCRIPTION
Correctness	Does the schema follow proper relational design principles? Are tables correctly structured with appropriate constraints and keys?
Completeness	Are all six validation tests passing? Does the database contain sufficient realistic data to return meaningful results?
Decision Quality	Did you correctly identify entities from the Excel dictionary? Are your design decisions clearly justified?
Documentation	Is your thought process clearly written and honest? A schema with minor issues but excellent documentation scores higher than a perfect schema with no reasoning.

09 SUBMISSION

- SQL script (.sql) — complete DDL and DML in a single file
- Written documentation as a Word or PDF document covering all requirements in Section 07
- Screenshots of all six validation queries embedded in your documentation

Submit to	datawithdany@gmail.com
Subject line	DWD SQL Assignment — [Your Full Name] — Cohort 8
Deadline	Wednesday — 2 weeks from issue date

A Note Before You Start

The Excel file has problems. Some column names are misleading. Some values are stored in formats that cannot be used directly in a database. Some columns contain data that belongs in multiple tables. That is intentional.

Real operational data is never clean. Your job is to make decisions in the face of that ambiguity, document why you made them, and build something that actually works. The thinking matters as much as the output.

Data With Danny | datawithdany@gmail.com | @iam_daniell | DWD-SQL-2026-001